



Challenge the Statistics

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Statistical Inference and Asking Tough Questions

"Oh, people can come up with statistics to prove anything, Kent. Forty percent of all people know that."

The Simpsons, Homer Simpson

What We'll Cover

- What is statistical inference?
- Hypothesis testing (null vs. alternative)
- p-values and significance levels
- Type I and Type II errors
- The process of statistical inference
- Questions to challenge statistics
- Statistics help us infer what's true from limited data.

The Big Idea – From Sample to Population

- We can't measure everything
- We use samples to understand the world
- Statistical inference helps us make informed guesses

"Statistical inference enables us to sample data from the world we live in to then make informed guesses about that world."

Why Challenge Statistics?

- Statistics can be misleading
- Numbers aren't always truth
- Blind acceptance leads to bad decisions

"Have you ever heard a statistical claim in the news or in the workplace that you wanted to understand, appreciate, or perhaps challenge? That's what this chapter is about."

The Role of a Data Head

- Question the numbers you see
- Understand the assumptions
- Ask for context and details
- Don't accept statistics at face value

Being a Data Head means being a critical consumer of statistics.

Give Yourself Wiggle Room

Poll: 655 out of 1,000 students said "yes"

65.5% is a point estimate

Would you say exactly 65.5%? Or want some wiggle room?

Confidence interval: $65.5\% \pm 3\% = (62.5\%, 68.5\%)$

"You probably wanted some wiggle room. That's good."

Definition: A confidence interval gives a range of plausible values.

More Data, More Evidence

- Amazon product with 1-star rating based on 1 review → ignore it
- Same product with 300 reviews → believe it
- **N = 300 is more trustworthy than N = 1**
- Sample size plays an outsized role in statistical inference

"This shows you're already accustomed to understanding how the number of data points behind a statistic... affects your trust in it."

Challenge the Status Quo

- Status quo = Null Hypothesis (H_0)
- New knowledge = Alternative Hypothesis (H_a)
- Burden of proof is on data to show status quo is wrong

Analogy: Innocent until proven guilty

- Defendant is assumed innocent (null)
- Evidence must prove guilt beyond reasonable doubt

"Fundamentally, science and the creation of new knowledge is about challenging the status quo."

Hypothesis Test Examples

Question: Has customer satisfaction changed?

- Null: Satisfaction has NOT changed
- Alternative: Satisfaction HAS changed

Question: Was the Super Bowl ad effective?

- Null: Ad did NOT change profits
- Alternative: Ad DID change profits

Question: Does the vaccine work?

- Null: Vaccine is NO better than placebo
- Alternative: Vaccine IS better

"Researchers want to find evidence to reject the null in order to support the alternative."

Evidence to the Contrary – The Basketball Intern

Intern claims he makes $\geq 50\%$ of shots (null hypothesis)

Game starts... he misses 10 shots in a row

What's the probability of 10 misses in a row?

$$0.5^{10} = 0.00098 \approx 1 \text{ in } 1,000$$

This probability is the **p-value**

"That's strong enough evidence that the original claim is unlikely to be so."

What is a p-value?

p-value = probability of seeing this result (or more extreme) **given the null is true**

Small p-value = evidence against the null

Common threshold (significance level) = 0.05 (5%)

If $p\text{-value} < 0.05$, we reject the null

This is called **statistically significant**

"If the p-value is less than the significance level, you reject the null and say the result is statistically significant."

Decision Errors; Type I and Type II

Type I Error (False Positive):

- Rejecting a true null hypothesis
- "Sending an innocent person to jail"
- $P(\text{False Positive}) = \text{significance level}$

Type II Error (False Negative):

- Failing to reject a false null hypothesis
- "Letting a guilty person go free"
- $P(\text{False Negative}) = 1 - \text{Power}$

"When variation leads to the wrong conclusion, this is called a decision error."

The Tradeoff ;You Can't Reduce Both

- Unless you collect more data, you cannot decrease one without increasing the other

Examples:

- **Spam Filter:** Low false positives (mom's email not spam) → more spam in inbox (more false negatives)
- **Disease Screening:** Accept more false positives → decrease false negatives (missed diagnosis)

"False positive and false negative errors have inherent tradeoffs."

The Intern Example Revisited

Question: Did intern's work improve customer satisfaction?

Null: This quarter's rate $\leq$ last quarter's rate

Alternative: This quarter's rate $>$ last quarter's rate

Data:

Last quarter: 50/100 recommend

This quarter: 65/100 recommend

p-value = 0.02 < 0.05

Conclusion: Statistically significant improvement

But wait... Is this causation or just correlation?

The "Significant" Trap; Statistical $\neq$ Practical

- Statistical significance doesn't mean practical significance
- Small effects can be found with large samples
- Large effects can be missed with small samples
- Always ask about effect size
- What's the magnitude of the difference?
- Does it matter to the business?

"Trivially small effects can be found with large sample sizes."

The Process of Statistical Inference ; Six Steps

1. Ask a meaningful question
2. Formulate hypotheses (null and alternative)
3. Establish significance level (usually 0.05)
4. Calculate the p-value
5. Calculate confidence intervals
6. Reject the null if $p\text{-value} < \text{significance level}$

"If you can read and understand those six steps, congratulations! You are learning the language of statistics."

Confidence Intervals Explained

- Confidence level = $1 - \text{significance level}$
- 5% significance $\rightarrow$ 95% confidence

Example:

- Poll with $N=1,000$: (62.5%, 68.5%) $\rightarrow$ narrow
- Poll with $N=100$: (54.8%, 74.2%) $\rightarrow$ wide
- More data = narrower interval
- If $N = \text{ALL}$ (census), no interval needed

"Confidence intervals also provide estimate of effect sizes in a statistical test."

Significance vs. Confidence; Two Sides of the Same Coin

"We rejected the null with 95% confidence"

"We rejected the null at $\alpha = 0.05$ "

These statements mean the same thing

Confidence level = 1 – Significance level

Always ask to see the confidence intervals, not just p-values

"This is why you should never be satisfied with just p-values."

The Data Head's Checklist; Overview of Questions

1. What is the context for these statistics?
2. What is the sample size?
3. What are you testing?
4. What is the null hypothesis?
5. What is the significance level?
6. How many tests are you doing?
7. Can I see the confidence intervals?
8. Is this practically significant?
9. Are you assuming causality?

"We created the following list of questions and prompts to ask your teammates to help you confidently challenge the statistics presented to you."

context

Context is just as important as the numbers

"Sales are up 10%" → Compared to what?

Example:

YouTube ad click-through rate improves by 50%

Context: 0.1% → 0.15% (absolute increase of 0.05%)

Relative change: 50%

Absolute change: 0.05%

"If you hear, 'Sales are up 10%' your brain should reflexively think, 'Compared to what?'"

Sample Size

Small N = high variation

Large N = more reliable

But:

N = ALL is rare

Big data $\neq$ all data

Excluding voices is a problem

"The assumption we make that N = ALL is one of the biggest problems we face in the age of Big Data. It is, above all, a way of excluding the voices of people who don't have the time, energy, or access."

What Are You Testing?

Every statistic has a question behind it

Don't let people share numbers without sharing the question

Make sure the question is clear and non-technical

"Don't let a data worker share numbers without sharing the underlying question."

Null Hypothesis

The null is the status quo

Science challenges the status quo

The alternative should be what you hope to prove

The Intern's Mistake:

Null: "This quarter's rate $\geq$ last quarter's rate"

This sets him up for victory by default

Correct Setup:

Null: "This quarter's rate $\leq$ last quarter's rate"

Alternative: "This quarter's rate $>$ last quarter's rate"

"A poorly defined null can create the deception where something is right simply because no data exists to show it's wrong."

Significance Level

Conventional = 5% or 0.05

Some use 1% (e.g., CERN physics research)

Lower significance = higher bar to reject null

But: Lower α = fewer false positives = more false negatives

Warning:

Don't set α after seeing the p-value

"Throwing a dart and then moving the target"

"A 5% significance means you can tolerate that you will wrongly reject a null hypothesis 1 out of 20 times."

How Many Tests?

Running many tests at 5% significance

100 tests → 5 will be significant by chance alone

Example:

Testing 100 foods against cancer

At $\alpha=0.05$, 5 foods appear significant by luck

Even if NO foods have a real effect

"With 100 different types of food in the database and a 5% significance level, 5 foods would appear as statistically significant in the fight against cancer even if no foods have a true effect."

Confidence Intervals

Confidence interval = range of plausible values

Shows uncertainty in the estimate

Wider interval = more uncertainty

Basketball Example:

U.S. players mean height = 72 inches

European players mean height = 71.5 inches

Difference = 0.5 ± 0.4 inches

"Statistical significance is often viewed as the seal of approval... But statistical tests are looking for any difference, whether important or not."

Practical Significance

Statistical significance $\neq$ practical significance

Large sample sizes find tiny effects

A 0.5-inch height difference is statistically significant but practically meaningless

Ask:

Does this result matter to the business?

Is the effect size meaningful?

What's the real-world impact?

"Trivially small effects can be found with large sample sizes."

Causality

Correlation $\neq$ Causation

Observational data can't prove causality

Only experiments (randomized controlled trials) can

The Intern's Mistake Again:

Customer satisfaction improved

Intern claims credit

But many factors could have caused the improvement

Unless there was a designed experiment, no causality

"Correlation is not causation. Unless there was a designed experiment... then you do not have causality."

Your Challenge Checklist; One Page Reference

1. **Context:** Compared to what?
2. **Sample Size:** How many observations?
3. **Testing:** What question is being answered?
4. **Null Hypothesis:** What's the status quo?
5. **Significance Level:** What's the threshold?
6. **Multiple Tests:** How many are being run?
7. **Confidence Intervals:** What's the range?
8. **Practical Significance:** Does it matter?
9. **Causality:** Are you assuming cause and effect?

"Armed with this list, you'll be able to effectively challenge, understand, and appreciate the statistics you see."

Statistical inference helps us make sense of the world

It requires asking tough questions

Don't accept numbers at face value

Be a skeptical, but informed, consumer of statistics

Key Takeaways:

Variation complicates decision making

Statistical significance $\neq$ practical significance

Correlation $\neq$ causation

Context matters

Ask the right questions

"Statistical inference is about using data to make better decisions, not about finding absolute truth."